MigrateAI

Intelligent Data Migration Assistant

Move your data anywhere. No pipelines. No scripts. No waiting.

A conversational AI platform that turns a 2-week IT ticket into a 5-minute chat — purpose-built for enterprise data teams.

Prepared for
Client Presentation

Classification
Confidential

Status
Proof of Concept

SCROLL

01

THE PROBLEM

Data migration is one of the most common — and most costly — enterprise bottlenecks.

Weeks of Waiting

01

Every data migration request flows through IT. The average turnaround from request to delivery is 2–3 weeks — regardless of complexity.

Avg. 2–3 weeks per request

Engineer Dependency

02

Business analysts, actuaries, and finance teams know exactly what data they need — but cannot move it themselves without a developer.

100% engineer dependency

No Tool for One-Time Loads

03

Fivetran and Airbyte serve ongoing pipelines. The majority of enterprise migration work — one-time historical loads — has no dedicated tooling.

~70% of requests are one-time

The result: a growing IT backlog, frustrated business teams, and engineers spending capacity on low-value data movement instead of high-value engineering work. **This is a solvable problem.**

THE SOLUTION

Introducing MigrateAI

💬

For Business Teams

Describe what you need in plain English. MigrateAI asks the right questions, collects the configuration, and executes the migration — no technical knowledge required.

⚡

For Data Engineers

Review and approve migration requests without spending time on requirements gathering. One click to execute. Full validation report delivered automatically.

🏢

For the Organisation

A self-service data movement capability that reduces IT backlog, accelerates decision-making, and scales without additional headcount.

"No code. No pipelines. No forms. **Just a conversation.**"

03

THE PROCESS

Three steps from request to result

1

Describe

The user opens MigrateAI and describes what they need in plain English. The AI asks targeted follow-up questions — only what is needed for that specific migration path.

NATURAL LANGUAGE

2

Configure

MigrateAI collects source details, destination details, filters, and preferences through conversation. No form to fill. No documentation to read.

AI-DRIVEN CONFIG

3

Execute & Validate

With a single confirmation, MigrateAI runs the migration, validates row counts pre and post transfer, and delivers a plain-English summary to all stakeholders.

AUTO-VALIDATED

4	4	<5m	0
Source connectors	Destination connectors	Time to migrate	Lines of code required

04

CURRENT STATUS

A working proof of concept — demonstrated today

Conversational config collection	✓ Complete
GPT-4o agent with tool calling	✓ Complete
4 source connectors (Postgres, CSV, S3, MongoDB)	✓ Complete
4 destination connectors (BigQuery, SQLite, S3, Snowflake)	✓ Complete
Pre and post migration validation	✓ Complete
Quick-start migration paths	✓ Complete
Branded UI — Liberty Mutual palette	✓ Complete
Multi-framework UI (Gradio, Streamlit, React + FastAPI)	✓ Complete
Session isolation — one agent per user	✓ Complete
Docker deployment configuration	✓ Complete

This is not a concept. It is a working system that can be demonstrated live in this session.

LIVE DEMONSTRATION

See it in action

1. User opens MigrateAI branded chat interface →
2. User clicks "Postgres → BigQuery" quick start →
3. AI asks targeted questions — host, database, table, destination →
4. User confirms — migration executes with validation →
5. Summary delivered — rows transferred, destination confirmed

📋 **For the presenter**

The live application is running and ready. Walk through one complete migration from the Quick Start click through to the validation summary. Estimated time: 3–4 minutes. The "Last Migration" panel updates in real time as the migration completes.

06

TECHNICAL OVERVIEW

Built on a clean, extensible architecture

CONVERSATION

Conversation Layer

GPT-4o powered agent. Collects configuration through natural dialogue. Validates completeness before executing. Explains results in plain English.

OpenAI GPT-4o · Tool Calling API · Python · FastAPI

MIGRATION

Migration Layer

Modular migration handlers. One handler per source-destination pair. Pre and post validation built in. Output verified and saved.

Python · Pandas · SQLAlchemy · Extensible connectors

INTERFACE

UI Layer

Three parallel implementations. Gradio for internal demos. Streamlit for business users. React + FastAPI for production deployment.

React · Streamlit · Gradio · Docker

Key architectural principle: The conversation layer and migration layer are entirely independent of the UI. Adding a new interface or a new connector requires no changes to the core system.

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THE PATH FORWARD

From proof of concept to enterprise platform

NOW — COMPLETE

Foundation

- Working POC with 4 sources + 4 destinations
- Conversational AI config collection
- Pre/post validation
- Multi-framework UI deployment
- Docker containerisatio

MONTHS 3–4

Production Hardening

- Real database connectors
- Secrets manager integration
- Full audit trail
- Error recovery + retry
- On-premise LLM option

MONTHS 5–6

Enterprise Features

- Request approval workflow
- SSO + role-based access
- Schema comparison
- ServiceNow / Jira integration
- Scheduled migrations

MONTH 7+

Scale

- 20+ source connectors
- Self-service connector builder
- Analytics dashboard
- Programmatic API access
- Multi-tenant deployment

MARKET TIMING

Three forces make this the right moment

01

LLM tool-calling is production-ready

The conversational configuration at the heart of MigrateAI was not reliably possible 18 months ago. GPT-4o's structured tool-calling makes it dependable enough for enterprise use.

02

Enterprise data is more fragmented than ever

Post-acquisition integrations, cloud migrations, and legacy modernisation programmes are creating more one-time migration demand than at any point in the past decade.

03

Self-service is the new expectation

Business users expect to accomplish tasks without filing tickets. Salesforce, Notion, and Figma have set the standard. Data movement should be no different.

COMPETITIVE CONTEXT

MigrateAI fills the gap enterprise ETL was never designed for

	Enterprise ETL Informatica · Talend · Azure Data Factory	MigrateAI
Designed for	Data engineers	Business users + data teams
Setup time	Weeks to months	Under 10 minutes
Interface	Complex visual designer	Plain English conversation
Best suited for	Ongoing pipelines, CDC, complex transforms	One-time loads, ad-hoc migrations
Annual licence cost	$50,000–$500,000+	Fraction of the cost
Skills required	Certified ETL developers	Anyone who can type
Time to first migration	Days to weeks	Under 10 minutes
CDC / incremental sync	✓ Core capability	Out of scope by design

Enterprise ETL solves the 20% of scenarios that represent 80% of complexity. **MigrateAI solves the other 80%** — the everyday, ad-hoc, one-time requests that enterprise ETL is too expensive and too slow for. These two tools are complementary, not competitive.

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REAL-WORLD APPLICATION

Where MigrateAI creates immediate value

POST-ACQUISITION

Data Consolidation

When a company acquires another organisation, historical data must move from the acquired company's systems into the parent warehouse before any analysis can begin. Today this takes weeks of engineering time.

→ Integration lead self-serves in hours, not weeks

ACTUARIAL

Model Data Feeds

Actuaries require fresh claims, exposure, and reinsurance data for pricing models. Getting this into a usable format currently requires either a scheduled pipeline or a manual engineer extract.

→ Actuary requests data themselves — no ticket filed

COMPLIANCE

Regulatory Reporting Prep

NAIC filings, state submissions, and Solvency II reporting require policy data extracted and formatted precisely. Today this involves manual extraction and transformation for every submission cycle.

→ Full audit trail included, compliance team self-sufficient

IT MODERNISATION

Legacy System Retirement

Retiring a legacy policy admin platform requires years of historical data preserved in the new environment. The historical preservation piece — moving flat tables between systems — is exactly what MigrateAI handles.

→ Reduces consulting scope for historical data preservation

The common thread: **the person who knows what they need is not the person who can get it today.** MigrateAI closes that gap.

THE BUSINESS CASE

Quantifiable value from day one

TIME PER MIGRATION REQUEST

3 wks

↓

<30m

95% reduction in time from request to delivery

ENGINEERING HOURS PER MONTH

160h

↓

0h

160 hours returned to high-value engineering work monthly

MIGRATION ERROR RATE

Manual

↓

Auto

Every migration validated pre and post — errors caught automatically

Risk Reduction

Manual migrations are error-prone. MigrateAI's pre and post validation catches discrepancies automatically. Every migration is logged, auditable, and reproducible — meeting enterprise compliance requirements.

Scale Without Headcount

As the organisation's data needs grow, MigrateAI scales without adding engineers. The bottleneck shifts from execution capacity to business demand — the right constraint to have.

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NEXT STEPS

Three things to move forward

1

Validate Priority Use Cases

Which migration workflows are causing the most pain today? We would like to spend time with the data team and key business stakeholders to identify the top three use cases for Phase 2 focus.

2

Access to a Test Environment

To demonstrate real connector capability, we need read access to one non-production source and write access to one non-production destination — a staging database and a sandbox environment is sufficient.

3

Architecture Alignment

A working session with the security and architecture teams to establish the credential management model, deployment architecture, and data governance requirements for the production path.

Proposed timeline: Live demonstration today → Working session with data engineering team within two weeks → Phase 2 scope definition and sign-off within four weeks.

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Data movement should be as simple as having a conversation.

MigrateAI is not a concept. It is a working system, built on a production-ready architecture, with a clear path from proof of concept to enterprise deployment.

Download this presentation →

Classification	Version	Technology	Status
Confidential	1.0 — POC Review	GPT-4o · Python · React	Live & Demonstrable